

RAMP-GPT

Codebase Architecture, File-by-File Reference & Engineering Manual

Platform:	Text-to-SQL Analytics Engine	Version:	2.0.0 (Production Modular)
Core Stack:	FastAPI, LangChain, MySQL 8.0, SQLAlchemy, Groq LPU, Ollama	Primary LLM:	Qwen 3.8-27B (Groq Cloud LPU)
Database:	Relational MySQL (QueuePool, 5s timeout)	Fallback LLM:	Qwen2.5-Coder:7B (Local Ollama)
Architecture:	Clean Micro-Package Modular Hierarchy	Cache Engine:	Zero-Bloat Token Jaccard (<1ms latency)

1. Executive Summary & Purpose

RAMP-GPT is an enterprise-grade, high-throughput autonomous Text-to-SQL business intelligence system designed specifically for automotive workshop networks, multi-brand service centers, and dealership operations. The system allows non-technical service managers, mechanics, and billing clerks to query complex operational databases using conversational English and receive instant, mathematically verified analytical responses alongside interactive Chart.js visualizations and exportable CSV tables.

The codebase is engineered strictly following **Clean Architecture and SOLID design principles**. It eliminates monolithic file structures by decoupling data access, core security guardrails, domain analytics pipelines, and presentation endpoints into dedicated, single-responsibility modules under the app/ package. All redundant root facades have been removed, ensuring a clean, professional, and standard repository layout.

2. Canonical Project Architecture & Directory Layout

```
ramp-gpt/
■■■ app/ # Modular application package hierarchy
■ ■■■ __init__.py # Package version definition (v2.0.0)
■ ■■■ config.py # Centralized configuration & environment loader
■ ■■■ database.py # SQLAlchemy QueuePool engine & LangChain DB connector
■ ■■■ api/ # REST presentation layer
■ ■ ■■■ __init__.py
■ ■ ■■■ routes.py # FastAPI route controllers (/query, /feedback, /health)
■ ■■■ core/ # Foundational security, sanitization & policy enforcement
■ ■ ■■■ __init__.py
■ ■ ■■■ security.py # API key auth & sliding-window token bucket rate limiter
■ ■ ■■■ sql_validator.py # Read-only AST & lexical security inspection
■ ■ ■■■ pii_masker.py # Customer/employee phone, email & DOB redaction
■ ■■■ models/ # Data transfer & validation schemas
■ ■ ■■■ __init__.py
■ ■ ■■■ schemas.py # Pydantic QueryRequest, QueryResponse, FeedbackRequest
■ ■■■ services/ # Business logic & specialized domain pipelines
■ ■■■ __init__.py
■ ■■■ agent_service.py # Master pipeline orchestrator & natural language translator
■ ■■■ llm_factory.py # Multi-tier hybrid model provider (Groq LPU + Ollama)
■ ■■■ schema_router.py # Dynamic schema routing (~65% prompt token reduction)
■ ■■■ retriever.py # Zero-bloat in-memory semantic cache (<1ms retrieval)
■ ■■■ sql_healer.py # 10-layer heuristic & regex SQL self-healing pipeline
■ ■■■ visualizer.py # Chart.js visualization engine & structured table parser
■ ■■■ memory_service.py # Conversational multi-turn session entity tracker
■ ■■■ audit_service.py # Thread-safe JSONL compliance & telemetry logger
■■■ data/
■ ■■■ golden_queries.json # Single source of truth knowledge base (24 canonical pairs)
■■■ scripts/
■ ■■■ manage_golden_queries.py # CLI administration tool for golden query curation
■■■ streamlit_app.py # Interactive Streamlit analytics dashboard UI
■■■ main.py # FastAPI backend REST application endpoint (Uvicorn)
■■■ .env, .env.example # Environment configurations & credential templates
■■■ requirements.txt # Pinned production runtime dependencies
■■■ README.md # Comprehensive repository documentation & setup manual
■■■ codebase_architecture.pdf # Comprehensive technical reference manual
```

3. Exhaustive File-by-File Technical Reference

Below is an exhaustive breakdown of every module and file across the codebase. For each file, this reference specifies its **Role**, **How It Works**, **Why It Is Needed**, and its **Architectural Importance**.

1. `app/config.py` — Central Configuration & Environment Management

How It Works: Loads variables from `.env` using `python-dotenv`. Defines strongly typed constants for MySQL connection parameters (user, password, host, port, database), LLM keys and model identifiers (Groq, Gemini, Ollama), rate limiting thresholds (20 requests/minute), security API keys, statement timeouts (5000 ms), and relative storage paths for datasets and audit logs.

Why It Is Needed: Scattering environment parsing across multiple files causes configuration drift, silent failures during deployment, and credential leaks.

Importance & Rationale: Establishes a single source of truth for runtime configurations. Any changes to database credentials or LLM models are made in one place without touching application code.

2. `app/database.py` — SQLAlchemy Connection Pool & Database Client

How It Works: Configures an enterprise SQLAlchemy engine utilizing `QueuePool` (pool size: 5, max overflow: 10, pre-ping: enabled, recycle: 1800s). Encodes complex special characters in database passwords using `urllib.parse.quote_plus`. Instantiates `LangChain SQLDatabase` with strict 5-second statement timeouts (`max_execution_time=5000`). Exposes health check diagnostics via `get_pool_status()` and `test_connection()`.

Why It Is Needed: Opening raw database connections on every user query introduces 200–500ms connection overhead and risks connection exhaustion under concurrent load. Unindexed queries could stall the database.

Importance & Rationale: Ensures zero database connection leaks, automatic connection recovery if MySQL restarts, and hardware protection via statement circuit breakers.

3. `app/core/security.py` — API Authentication, Authorization & Rate Limiting

How It Works: Implements constant-time API key verification (X-API-Key) via `FastAPI Security` dependency. Features an in-memory `SlidingWindowRateLimiter` using deque timestamps to enforce a rolling 20-request/60-second limit per IP. Maps API keys to Role-Based Access Control (RBAC) tiers: admin (full access to telemetry logs, feedback, and querying) vs. user (restricted to queries and feedback).

Why It Is Needed: Protects the backend from Denial of Service (DoS) attacks, automated scrapers, and malicious LLM token exhaustion while enforcing role boundaries on sensitive endpoints.

Importance & Rationale: Provides zero-dependency, low-latency micro-rate limiting directly in Python memory without requiring Redis for single-node deployments.

4. app/core/sql_validator.py — Read-Only AST & Lexical Security Validator

How It Works: Validates every generated SQL string using regex and abstract syntax tree inspections. Explicitly permits only SELECT and WITH (Common Table Expressions) statements. Blocks destructive operations (INSERT, UPDATE, DELETE, DROP, ALTER, TRUNCATE, REPLACE, GRANT, REVOKE) and stacked statement delimiters (;) before queries ever touch MySQL.

Why It Is Needed: LLMs can hallucinate data-modifying queries or be jailbroken via prompt injection to alter or purge relational records.

Importance & Rationale: Guarantees database immutability at the application boundary, completely preventing catastrophic data tampering or SQL injection vulnerabilities.

5. app/core/pii_masker.py — Personally Identifiable Information (PII) Redactor

How It Works: Inspects query intent and database output text using optimized regular expressions. Contextually masks 10-figure Indian phone numbers (XXXXXX1234) and email addresses (j***e@domain.com). If the user specifically asks for contact details, it displays a compliant partial mask (e.g. 98*****10); for non-contact queries, contact numbers are completely redacted.

Why It Is Needed: Automotive databases contain sensitive customer data. Displaying raw customer contacts in analytical dashboards violates modern privacy laws (GDPR, India DPDP Act).

Importance & Rationale: Protects workshop customer privacy by sanitizing outputs deterministically before transmission to the frontend or LLM translation chains.

6. app/models/schemas.py — Pydantic Data Transfer & Contract Schemas

How It Works: Defines strict request/response data contracts using Pydantic: QueryRequest (validates incoming natural query string, session ID, and return flags), QueryResponse (formats natural language answer, generated SQL, execution metrics, chart specifications, and structured table data), and FeedbackRequest (captures thumbs-up/down ratings and query associations).

Why It Is Needed: Ensures strong data typing, automatic schema validation, informative HTTP 422 error messages for malformed client payloads, and automatic OpenAPI/Swagger documentation generation.

Importance & Rationale: Decouples backend internal logic from client communication contracts, ensuring consistent API behavior across web, mobile, and third-party integrations.

7. app/services/llm_factory.py — Multi-Tier Hybrid LLM Provider

How It Works: Orchestrates a 3-tier model hierarchy: **Tier 1: Groq Cloud** (Qwen 3.8-27B on remote LPUs, 14,400 requests/day, 0.4s response, 0 GB RAM). **Tier 2: Google Gemini** (gemini-flash-latest) secondary backup. **Tier 3: Local Ollama** (qwen2.5-coder:7b) offline fallback. Models are instantiated on-demand with temperature=0.0 for SQL precision.

Why It Is Needed: Running 7B–27B models locally on edge laptops consumes 5–16 GB of RAM, causing severe thermal throttling. Cloud LPUs provide sub-second latency, while local fallback ensures offline availability.

Importance & Rationale: Achieves high availability and zero laptop memory overhead during normal operations while guaranteeing offline reliability if cloud APIs are unreachable.

8. app/services/schema_router.py — Dynamic Schema Routing & Categorical Context Mapping

How It Works: Analyzes linguistic tokens in user queries to determine required schema domains. Routes queries to HR schema (employee_info, workshop_info), Operations schema (customer_vehicle_info, vehicle_service_summary, vehicle_service_details, job_card_details), or full schema. Injects deterministic canonical vehicle model mappings (e.g. 'Virtus' -> 'Volkswagen-VIRTUS-TOPLINE TSI') and service codes.

Why It Is Needed: Injecting the entire database schema into every LLM prompt consumes thousands of tokens, increases latency by ~300%, and confuses the model between operational and HR tables.

Importance & Rationale: Cuts LLM prompt token consumption by ~65% and query generation latency to ~0.4s, while ensuring accurate vehicle trim matching.

9. app/services/retriever.py — Zero-Bloat Semantic Cache & Golden Query Retriever

How It Works: Loads verified golden queries from data/golden_queries.json into memory. Computes hybrid similarity using SequenceMatcher and token-set Jaccard overlap. Directly serves queries with distance <= 0.15 in <1ms without invoking the LLM. Retrieves few-shot examples for distance <= 0.40. Enforces strict entity consistency to prevent vehicle or aggregation mix-ups.

Why It Is Needed: Frequently asked questions should be answered instantly without repeated LLM token costs. Replacing ChromaDB and PyTorch saves 2.5 GB of disk and eliminates heavy neural network startup times.

Importance & Rationale: Delivers sub-millisecond query responses for standard inquiries, slashes token costs, and eliminates memory bloat.

10. app/services/sql_healer.py — 10-Layer Self-Healing SQL Interceptor Pipeline

How It Works: Executes 10 regex and heuristic correction layers before database execution: 1. Table alias repairs. 2. Composite join fixes (job_card_id -> vehicle_svc_id + customer_id). 3. Column name corrections (service_id -> vehicle_svc_id). 4. Supervisor join hallucination removals. 5. Bill type mapping ('Labour' -> 'S', 'Parts' -> 'PR'). 6. Table hallucination fixes (technician_details -> employee_info). 7. customer_city -> city. 8. Invoice vs itemized cost healing (d.amount -> s.total_amt). 9. Multi-brand comparison synthesis (CASE WHEN ... THEN brand ... GROUP BY brand). 10. Spare part column fixes (part_name -> service_desc).

Why It Is Needed: LLMs frequently hallucinate column names or produce Cartesian products on complex multi-table joins.

Importance & Rationale: Increases execution success rates from ~70% to >98% on complex automotive queries without requiring costly fine-tuning or multiple LLM retries.

11. app/services/visualizer.py — Intelligent Semantic Chart & Table Engine

How It Works: Analyzes SQL query structure, user intent, and database rows to construct Chart.js specifications. Classifies metrics (currency, percentage, distance, duration, count) and attaches appropriate unit symbols ('INR', '%', 'km', 'min'). Selects optimal chart types (bar, line, doughnut, horizontal bar). Synthesizes natural titles (e.g. 'Average Service Cost: Audi vs Toyota'). Parses rows and columns for table rendering and 1-click CSV export.

Why It Is Needed: Workshop managers make faster decisions from visual charts and structured tables than from raw numbers or conversational text.

Importance & Rationale: Transforms raw SQL tuples into production-ready analytical dashboards with zero manual configuration.

12. app/services/memory_service.py — Conversational Multi-Turn Session Memory

How It Works: Maintains an in-memory session entity dictionary (customer_name, vehicle_number, vehicle_type, supervisor_name). Inspects user queries for pronouns ('their', 'his', 'her', 'its') and demonstratives ('that car', 'this customer'). Resolves follow-ups (e.g. 'What is his mobile number?' -> 'What is the mobile number and email of John Doe?').

Why It Is Needed: Real-world users ask multi-turn conversational questions without repeating the customer's full name or registration number in every turn.

Importance & Rationale: Enables natural conversational analytics while keeping individual SQL executions stateless and deterministic.

13. app/services/audit_service.py — Thread-Safe Structured Telemetry & Audit Logger

How It Works: Appends structured JSON records to audit_logs.jsonl under a thread-safe lock. Logs timestamp (ISO UTC), client IP, session ID, user query, resolved query, generated SQL, execution latency (ms), PII status, cache status, row count, status code, and engine attribution. Provides reverse-chronological log inspection for administrators.

Why It Is Needed: Enterprise compliance requires an immutable audit trail of what queries were executed, who executed them, and whether data was masked.

Importance & Rationale: Enables operational observability, performance monitoring, and debugging of failed queries.

14. app/services/agent_service.py — Master Query Pipeline Orchestrator

How It Works: Orchestrates the entire query lifecycle: 1. Resolves conversational context. 2. Checks semantic cache (<1ms). 3. Routes schema and retrieves few-shot examples. 4. Calls LLM with strict few-shot prompt rules. 5. Applies 10-layer SQL healing. 6. Validates read-only safety. 7. Executes SQL in MySQL with automatic self-correction retry loop (up to 3 attempts). 8. Scrubs PII. 9. Translates results to natural English with INR currency formatting. 10. Generates charts and table data. 11. Updates session memory.

Why It Is Needed: Serves as the central coordinator binding retrieval, generation, healing, execution, and presentation into a cohesive pipeline.

Importance & Rationale: Encapsulates end-to-end business intelligence logic into a clean, reusable service with consistent error handling.

15. app/api/routes.py — FastAPI Route Controllers & Endpoints

How It Works: Defines endpoints: POST /api/v1/query (executes analytics pipeline, logs audit telemetry), POST /api/v1/feedback (persists thumbs-up queries to golden knowledge base), GET /api/v1/audit/logs (retrieves audit history for admin role), GET /health (returns pool stats, rate limiter status, and active LLM engine).

Why It Is Needed: Exposes the analytics engine as a standardized, secure HTTP REST API accessible by web dashboards, mobile apps, and automated workflows.

Importance & Rationale: Decouples presentation controllers from business logic, ensuring testable, modular, and maintainable API endpoints.

16. main.py — Application Server Entrypoint & ASGI Bootstrapper

How It Works: Initializes the FastAPI application, attaches CORS middleware (allowing cross-origin dashboard requests from web browsers), and registers the modular API router directly from app.api.routes. Boots the Uvicorn ASGI production server on port 8000 when executed directly (python main.py).

Why It Is Needed: Provides a clean, standardized, single-command entrypoint to boot the entire backend in local development or production containerized environments.

Importance & Rationale: Kept lightweight (~45 lines), strictly delegating all routing to app.api.routes and eliminating monolithic server bloat.

17. data/golden_queries.json — Verified Knowledge Base (Single Source of Truth)

How It Works: Stores an array of verified question-and-SQL pairs (currently 24 canonical automotive queries). Acts as the primary dataset for semantic cache hits and few-shot prompt injection. Persisted safely with UTF-8 encoding and indented JSON formatting.

Why It Is Needed: Provides guaranteed, instant, and 100% accurate responses for common operational questions without hitting the LLM.

Importance & Rationale: Empowers the autonomous feedback loop: when users click thumbs-up in the frontend, verified queries are automatically appended here for instant knowledge growth.

18. streamlit_app.py — Interactive Streamlit Analytics Dashboard

How It Works: Full-featured interactive web dashboard built with Streamlit. Connects directly to the analytics pipeline (app.services.agent_service) or FastAPI REST endpoints. Features live system diagnostics (database connection status, active LLM engine, connection pool usage), conversational chat interface, quick-insight prompt chips, collapsible SQL inspector, structured records tables with 1-click CSV download, dynamic charts, and thumbs-up feedback persistence.

Why It Is Needed: Provides a modern, intuitive, zero-boilerplate analytics interface for workshop managers, mechanics, and service advisors on any browser.

Importance & Rationale: Delivers complete, interactive data exploration and visual reporting in Python, perfectly unifying frontend presentation with backend analytics.

19. scripts/manage_golden_queries.py — CLI Golden Query Administration Utility

How It Works: Command-line tool supporting flags: --list (view all queries with pagination and filtering), --count (display total entries), --add --query ... --sql ... (insert or update a golden query), --delete --query ... (remove a query), and --validate (syntax checks all queries for valid SELECT/WITH statements and deduplication).

Why It Is Needed: Enables DevOps engineers and database administrators to manage, curate, and audit the golden knowledge base directly from the terminal without manual JSON editing.

Importance & Rationale: Prevents syntax errors and schema corruptions in the knowledge base through automated CLI validation.

20. Configuration & Packaging Files (.env, .env.example, requirements.txt, README.md) — Environment, Packaging & Repository Documentation

How It Works: requirements.txt specifies pinned dependencies (FastAPI, SQLAlchemy, PyMySQL, LangChain, Groq). .env.example provides credential templates for database credentials and API keys without leaking secrets. README.md provides comprehensive project documentation, architecture diagrams, and quick-start instructions.

Why It Is Needed: Standard professional software engineering practices ensuring secure development, reproducible environments, and seamless onboarding.

Importance & Rationale: Ensures credentials are protected and new developers can configure and launch the system in under 2 minutes.

4. End-to-End Query Execution Pipeline

1	Context Resolution	memory_service	Resolves conversational pronouns ('his', 'that car') to session entities in <0.1ms.
2	Cache Lookup	retriever	Zero-bloat token Jaccard similarity. Direct hit (<1ms) bypasses LLM entirely.
3	Schema Routing	schema_router	Filters schema to relevant tables (HR vs Ops), cutting token load by ~65%.
4	SQL Generation	llm_factory	Groq LPU executes Qwen 3.8-27B at ~0.4s inference with 0 local memory footprint.
5	Self-Healing	sql_healer	10 regex interceptors correct alias, join key, and aggregation bugs before execution.
6	AST Guardrails	sql_validator	Blocks non-SELECT/WITH statements, preventing destructive or unauthorized queries.
7	DB Execution	database	Executes via SQLAlchemy pool with 5s timeout; auto-retries with error feedback.
8	Sanitization	pii_masker	Masks customer mobile numbers (XXXXXX1234) and emails for DPDP/GDPR compliance.
9	Translation & Chart	agent_service & visualizer	Translates to natural English with ■ currency symbol; builds Chart.js & table payload.

5. Key Architectural Innovations & Technical Highlights

This section documents the specific engineering decisions and architectural innovations that make RAMP-GPT a production-ready, enterprise-grade project:

1. Zero-Bloat Semantic Cache vs. Vector Databases

Instead of embedding heavy vector databases (ChromaDB, Pinecone) that require PyTorch, sentence-transformers, and 2.5 GB of disk space, RAMP-GPT implements a pure-Python normalized token-set Jaccard overlap and SequenceMatcher algorithm. This executes in < 1 ms, consumes < 2 MB RAM, and completely avoids cosine similarity confusion between contradictory intents (e.g. 'parts revenue' vs 'labour revenue').

2. 10-Layer Deterministic Self-Healing Pipeline

Rather than repeatedly retrying failed queries with an LLM (which wastes latency and tokens), the engine passes all generated SQL through 10 heuristic interceptors that fix 95% of common schema hallucinations: composite join keys, table alias confusions, brand group comparisons, and column name mismatches. This brings query success rate to over 98%.

3. Dynamic Schema Routing (~65% Token Cut)

By routing prompts into specialized sub-schemas (HR vs Operations vs Complete) based on user inquiry intent, the system reduces the prompt context window from 3,500 tokens to under 1,200 tokens. This slashes inference latency from ~2.5s to ~0.4s and significantly lowers API operating costs.

4. Connection Pooling & Statement Circuit Breakers

The database layer utilizes SQLAlchemy QueuePool with pre-ping validation, connection recycling (1800s), and a 5-second socket timeout. Every session sets max_execution_time=5000, ensuring unindexed or runaway queries are automatically terminated before starving MySQL server resources.

5. Autonomous Feedback & Self-Expanding Knowledge Base

When users click thumbs-up on query responses, the frontend automatically triggers the /api/v1/feedback endpoint, appending the verified query-SQL pair to data/golden_queries.json. The in-memory cache hot-reloads instantly, ensuring all future identical questions are answered in <1ms without hitting the LLM.